

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch :ENTC Semester: Vth Subject Code & Name:(BTEXC502) control system engineering Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following. (This is just a sample instruction)		12
A)	Explain open loop and closed loop control system with example.		06
B)	Explain block diagram reduction technique.		06
C)	Explain standard test input signals.		06
Q.2	Solve Any Two of the following. (This is just a sample instruction)		12
A)	Find overall gain $C(s) / R(s)$ for the signal flow graph		06
B)	Derive the expression for first order system for a unit step input.		06
C)	Define Control system with examples.		06
Q. 3	Solve Any One of the following. (This is just a sample instruction)		12
A)	Reduce the block diagram and find C / R .		12
B)	Sketch Root Locus for unity feedback system whose open loop transfer function is $G(s) = \frac{K}{s(s+4)(s^2+4s+20)}$		12

C)	Explain Routh –Hurwitz Criteria in detail.		12
Q.4	Solve Any Two of the following. <i>(This is just a sample instruction)</i>		12
A)	Explain controllability and observability of linear system.		06
B)	What are the advantages of state space analysis?		06
C)	Explain Gain margin and phase margin in detail.		06
Q. 5	Solve Any One of the following. <i>(This is just a sample instruction)</i>		12
A)	Trace polar plot and find gain margin and phase margin of $G(s) = \frac{1}{s(1+s)(1+2s)}$		12
B)	Test the observability of the system described by $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$ $Y = [1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$		12
C)	Determine the state transition matrix for the system $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$ Where $u > 0$		12
	*** End ***		

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